X-Theorem on Distinguishability and the Condition of Informational Existence

1 Introduction

In classical physics, objects are considered as entities existing in space and time and possessing a set of physical properties.

However, in an informational description, any entity is defined not only by its internal characteristics but also by a system of distinctions relative to other entities.

This work proposes a principle according to which distinguishability is a more fundamental concept than information, space, and observable dynamics, and the existence of an informational object is determined by the structure of its distinctions.

2 Basic Concepts and Notation

Consider a set of states:

$$S = \{s_i\}$$

where s_i are abstract states of a system.

A fundamental distinguishability relation is introduced:

$$\delta : S \times S \to \{0, 1\}$$

where:

- $\delta(s_i, s_j) = 1$ — states are distinguishable;
- $\delta(s_i, s_j) = 0$ — states are indistinguishable.

For an arbitrary object $X \subseteq S$, we introduce a measure of distinguishability:

$$D(X)$$

interpreted as the structure of distinctions separating $Xfromtherestofthesystemstates.$

3 Definition of Informational Object

An informational object X is defined as a structure of distinctions given by δ, rather than as an independent entity.

The existence of an object implies the distinguishability of its state relative to other states in the system.

4 Central Theorem (X-Theorem)

An informational object exists if and only if its structure of distinguishability is non-zero.

Mathematically:

$$X \text{ exists as an informational object} \iff D(X) > 0$$

$$D(X) = 0 \Rightarrow X \text{ does not possess informational identity}$$

5 Interpretation of the Result

- An object cannot be defined as an informational entity in the absence of distinctions.
- Informational existence requires a non-zero structure of distinguishability.
- Complete removal of distinctions leads to the disappearance of the object as an informational entity.

6 Example (Apple)

A physical apple:

Can be described as an object in space and allows limiting descriptions (including point-like mass). It remains a physical model.

An informational apple:

Exists through a system of distinctions:

- apple $\neq$ pear
- apple $\neq$ stone

- apple possesses a set of properties (shape, color, taste, etc.)

If all distinctions are removed, there is no "apple at a point" left — only the absence of an informational object.

7 Proof Basis (Structural Argumentation)

1. **Principle of distinguishability:** identification of an object is only possible if at least one distinguishing feature exists.
2. **Relational identity:** informational identity is defined through relations of difference, not intrinsic properties.
3. **Collapse of distinguishability:** when $D(X) = 0$, there is no mechanism to isolate the object as a distinct informational entity.

8 Corollary (Principle of Existence)

Informational existence requires a non-zero structure of distinguishability:

$$D(X) > 0$$

9 Conclusion

The X-Theorem formulates a minimal condition for the existence of an informational object.

Unlike physical objects, which allow limiting descriptions with zero localization, informational objects require a non-zero structure of distinguishability as a condition of their existence.

Thus, extending this principle to global system states leads to the interpretation that a completely homogeneous (undifferentiated) state is not physically realizable. This opens a pathway to consider the initial conditions of the universe as a state with a minimally non-zero structure of distinguishability, which in limiting form may be associated with the emergence of initial cosmological instability and, consequently, the Big Bang.

10 Central Formula

$$D(X) = 0 \Rightarrow X \text{ does not exist as an informational object}$$